Program : **Diploma in Civil and Rural Engineering**	
Course Code : **3367**	Course Title: **Advanced Survey Lab**
Semester :	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: 45

Course Objectives:

- To enable the students to perform contouring and to use theodolite for obtaining angular measurements and perform traversing.
- To enable students to perform contouring by Total station surveying
- To enable students to perform GPS surveying

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of Basic Mathematics		Engineering Mathematics	1
Basic Surveying		Basic Surveying	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Measure the horizontal and vertical angles using theodolite.	9	Applying
CO2	Perform traversing using theodolite and prepare plans and Height and distance by Tacheometric survey	12	Applying
CO3	Perform contouring using total station and prepare map of a plot	12	Applying
CO4	GPS surveying	8	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Measure horizontal and vertical angles using theodolite**		
M1.01	Perform temporary adjustments of theodolite and measurement of horizontal angle by repetition method	3	Applying
M1.02	Measure horizontal angles by reiteration method	3	Applying
M1.03	Measure vertical angle using theodolite	3	Applying
CO2	**Perform traversing using theodolite and prepare plans and Height and distance by Tacheometric survey**		
M2.01	Perform theodolite traversing and record observation.	3	Applying
M2.02	Compute included angles, latitude, departure, and prepare Gale's traverse table. Plot the traverse to find the area	3	Applying
M2.03	Determination of height and distance to point using stadia tacheometry	3	Applying
M2.04	Determination of height and distance to point using tangential tacheometry	3	Applying
	Lab test	2	

CO3	**Perform contouring using total station and prepare map of a plot**		
M3.01	Perform temporary setting of Total station	3	Understanding
M3.02	Perform measurement of horizontal and vertical distance, area and volume of the given plot	3	Applying
M3.03	Total station traversing and contouring	3	Applying
M3.04	Plotting of contour	3	Applying
CO4	**GPS surveying**		
M4.01	Utilize GPS to find out coordinates (longitude and latitude) of a station	3	Applying
M4.02	Open ended project	5	Applying
	Lab test	2	

Text / Reference:

T/R	Book Title/Author
T1	Punmia, B.C,; Jain, Ashok Kumar; Jain, Arun Kumar, Surveying I, Laxmi Publications., New Delhi
R2	Basak, N. N., Surveying and Levelling, McGraw Hill Education, New Delhi
R3	Kanetkar, T. P.; Kulkarni, S. V., Surveying and Levelling volume I, Pune Vidyarthi GruhPrakashan
R4	Duggal, S. K., Survey I, McGraw Hill Education, New Delhi.
R5	Saikia, M D.; Das. B.M.; Das. M.M., Surveying, PHI Learning, New Delhi
R6	Subramanian, R., Fundamentals of Surveying and Levelling, Oxford University Press. New Delhi
R7	De,Alak, Plane Surveying, S.Chand Publications, New Delhi.
R8	Rao, P.VenugopalaAkella, Vijayalakshmi, Textbook of Surveying, PHI Learning
R9	Venkatramaiah, C, Textbook of Surveying, Universities Press, Hyderabad.
R10	Anderson, James M and Mikhail, Edward M, Surveying theory and practice, Mc Graw Hill Education, Noida.

Online Resources:

Sl. No	Website Link
1	http://www.vlab.co.in/ba-nptel-labs-civil-engineering
2	https://nptel.ac.in/courses